IIT GANDHINAGAR • SSRP PROJECT 7 • STUDENT MASTER GUIDE

Controlling a 3D VR Weightlifting Game with Your Muscles!

From Bioelectric Sparks to 3D Virtual Reality: The Complete Story of EMG Data Collection, Real-Time Digital Signal Processing, and Modern Game Design.

HOLD & REPS MODE | 6 kg CALIBRATION | EMG • FILTER IIR | SCORE: 2,840 (x2)

Project Topic: Real-Time EMG Signal Processing for Biceps-Controlled VR Weightlifting
Platform: BioRadio Hardware • Python Data Bridge (UDP) • Unity 6 3D Engine
Target Audience: Complete A-to-Z explanation structured for easy, fun comprehension
Academic Context: Summer Semester Research Project (SSRP) • Bio-Signals & VR Lab
Date: November 2026

Chapter 1: The Magic of Muscle Electricity (What is EMG?)

Have you ever wondered how your arm moves when you want to lift a heavy school bag or throw a ball? It feels like magic, but your body is actually powered by **electricity**!

> **The Light Switch Analogy:**
> Imagine your brain is a game console, your nerves are electric cables, and your biceps muscle is a cluster of millions of tiny motors. When you decide to bend your elbow, your brain flips a switch and sends tiny electrical sparks (called *action potentials*) shooting down the wires into your muscle fibers!

When those sparks hit your muscle fibers, chemicals rush in, and the fibers pull tight against each other like microscopic tug-of-war teams. When millions of them pull at once, your biceps thickens, bunches up into a ball, and pulls your forearm upward.

What is Electromyography (EMG)?

If we stick small metal stickers on your skin directly over your biceps muscle, we can "eavesdrop" on those electrical sparks! That is what **EMG (Electromyography)** is: **Electro** (Electricity) + **Myo** (Muscle) + **Graphy** (Writing or recording) = "*Writing down the electrical activity of muscles*".

> **How Small is This Muscle Signal?**
> A normal battery for a toy is **1.5 Volts**. The electric signals from your biceps are tiny: only around **50 to 5,000 microvolts (µV)**! One microvolt is *one millionth* of a single volt. That means your muscle signal is **over 10,000 times weaker than a remote-control battery**! Listening to it is like trying to hear someone whispering softly in the middle of a roaring football stadium.

Chapter 2: Catching the Sparks (Data Collection & BioRadio)

To capture such microscopic electrical whispers without being drowned out by background noise, we use the **BioRadio**:

1. **The Sticky Electrodes (Ag/AgCl Stickers):** Two sticky silver/silver-chloride pads with conductive gel are placed on the belly of your biceps, 2 cm apart. A third pad (*ground reference*) is placed on a bony area like your elbow where no muscles flex.

2. **The BioRadio Device (The Super Amplifier):** The BioRadio box contains medical-grade instrumentation pre-amplifiers that multiply the voltage by thousands of times so the computer can measure it accurately.

3. **Sampling at 1,000 Hz (1,000 Snapshots Every Second!):** Human muscle electricity changes very quickly—up to 400–500 times a second. By the famous *Nyquist-Shannon rule*, we must capture at least double that speed: exactly **1,000 snapshots every single second**!

4. **Wireless Bluetooth Bridge:** Instead of dangerous wires while you exercise, the BioRadio packs these 1,000 numbers into radio packets and beams them over Bluetooth to the PC.

Chapter 3: The Noise Monsters (Why Raw EMG Looks Like Static)

If you look at raw EMG straight out of the BioRadio, you won't see a nice smooth curve. You will see a crazy, jagged, fuzzy scribble jumping wildly up and down! Here are the four culprits:

Monster 1: The 50 Hz Wall Hum

Mains Noise

Every wall outlet and light in the room hums with 50 Hz alternating current. Human bodies act like antennas that pick up this electrical buzzing.

Monster 2: Cable Sway & Skin Wobble

< 20 Hz

When you move, wires swing through the air and skin stretches under the stickers. This creates slow, giant rolling waves unrelated to muscle effort.

Monster 3: High-Frequency Radio Hiss

> 450 Hz

Wi-Fi signals, power supplies, and thermal vibrations create random high-pitched fuzz that contaminates the measurement.

Monster 4: The Zero-Sum Trap

Symmetry

Muscle action potentials are electrical dipoles: they shoot positive (+), then negative (-). If you average raw EMG, (+50) plus (-50) equals ZERO! A raw average says you did zero work even while straining!

Why We Can't Connect Raw EMG to the Game:

If you hooked raw EMG straight to the virtual arm, the arm would buzz like an angry bee 1,000 times a second, jumping between fully up and fully down, and then collapsing to zero. We must build a digital filter pipeline to tame the noise monsters first!

Chapter 4: The 4 Filter Superheroes & Envelope Extraction

To turn noisy scribbles into a clean muscle power number, our IIT Gandhinagar team built four digital filtering methods in `emg_methods.py`:

Method	Team Member	How It Works (12-Year-Old Explanation)	Superpower / Strength
IIR Filter (Infinite Impulse Response)	Raima	**The Smart Feedback Funnel:** To clean each incoming drop of data, it looks at the new dirty sample AND the clean samples that just flowed out. Uses a 4th-order Butterworth bandpass (20–450 Hz) and a 50 Hz notch filter.	**Lightning Fast!** Requires very few calculations. Has only ~40 milliseconds of delay, making it the perfect choice for real-time video games!
FIR Filter (Finite Impulse Response)	Kunal	**The Weighted Conveyor Belt:** Lines up the last 50 snapshots on a belt. Multiplies each by a calculated weight and sums them. Never feeds past outputs back into itself.	**Rock-Solid Stability:** Guaranteed to never blow up or ring, keeping all frequencies in perfect phase alignment.
FFT Filter (Fast Fourier Transform)	Chandan	**The Smoothie Separator:** Breaks the signal into pure musical frequencies (like separating a fruit smoothie back into bananas and milk), erases the 50 Hz hum notes, and glues the music back together.	**Surgical Precision:** Can completely delete exact unwanted noise frequencies from the spectrum.
DIFF Filter (Difference / Derivative)	Krishna	**The Speedometer:** Calculates how fast the signal is changing between sample A and sample B. Slow cable sway is erased because it changes very slowly.	**Instant Reaction:** Detects the very first millisecond a muscle begins to fire.

Turning Electricity into Muscle Effort: The Envelope Pipeline

1. **Step 1: Full-Wave Rectification (Absolute Value):** Flip all negative numbers into positive numbers: $|-120\,\mu\text{V}| = +120\,\mu\text{V}$. Now all electrical waves pull in the same positive direction!

2. **Step 2: Moving Average / RMS Smoothing (The Envelope):** Slide a 100-millisecond window across the rectified values and calculate the Root-Mean-Square (RMS). This draws a silky-smooth, continuous curve over the tops of all the spikes—called the **Linear Envelope**. When relaxed, it sits near zero; when flexing hard, it rises smoothly into a high hill!

Chapter 5: The "Mystery of the 6 kg Weight" & Two-Point Calibration

During testing, the team encountered a very strange bug:

> **The 6 kg Glitch:**
> When a player picked up the heavy 6 kg dumbbell and held their arm *completely straight down* without curling at all, the virtual game arm started lifting upwards on its own!

Why Did That Happen?

Many people thought: "*It's just noise! Let's make the filter window 500 milliseconds longer!*" But that was completely wrong. When holding a heavy 6 kg weight, your fingers squeeze tight to avoid dropping it, and your biceps tenses up to keep your arm in its socket. This produces a **real, steady electrical hum**. Because it is steady, no smoothing filter can remove it (smoothing filters only stop fast wiggles; steady levels pass right through). In the old game, any activation above `0.03` made the arm rise, so the 6 kg weight caused the virtual arm to float!

The Solution: Two-Point Calibration for Each Load

Instead of a one-size-fits-all rule, we teach the computer what *your* body looks like under both conditions:

1. Cal REST (5 Seconds)
Hold the load with your arm completely straight and relaxed for 5 seconds. The computer measures the 95th percentile of your resting signal. **This level becomes our defined ZERO (0.0).**

2. Cal MAX (8 Seconds)
Perform 3 full, strong curls with that weight within 8 seconds. The computer measures your peak flex energy (99th percentile). **This level becomes our defined ONE (1.0).**

> **The Calibration Formula:**
> $$\text{Activation} = \frac{\text{Current Envelope} - \text{REST}}{\text{MAX} - \text{REST}}$$ • Arm straight holding 6 kg: Envelope = REST → **Activation = 0.0 (Arm stays completely straight!)**
> • Full curl with 6 kg: Envelope = MAX → **Activation = 1.0 (Arm reaches the top!)**
> • **Dead Zone (5%):** Anything below 0.05 is clamped to exactly 0 to eliminate any resting finger twitches.

Chapter 6: The Secret Walkie-Talkie (Python to Unity via UDP)

Python calculates clean muscle power (0.0 to 1.0), and Unity displays the 3D arm bending. How do they communicate?

> **The Walkie-Talkie Protocol (UDP):**
> Python uses a high-speed message launcher called **UDP (User Datagram Protocol)**. Every 20 milliseconds (50 times every second!), Python writes a tiny note and shoots it across a private pipe inside Windows called **Port 50557** directly to Unity!

What Does the Secret Note Look Like?

The message is formatted in **JSON** (JavaScript Object Notation):

```
{ "channel": 1, "activation": 0.7240, // 72.4% flex power! "envelope_uv": 412.5, // Raw envelope in microvolts "filter": "IIR", // Using Raima's IIR filter "load": "6 kg", // Calibrated for 6 kg weight "sample_rate_hz": 1000.0, "sender_time": 1793997405.432 }
```

Why UDP instead of standard TCP? In video games, freshness matters most! If a packet gets dropped, we don't want the game to pause and wait; we only care about what your muscle is doing *right now*. UDP is blazing fast with zero lag.

Chapter 7: Building the 3D Game World & Character in Unity

How did we build the high-end 3D gym studio, the muscular lifter, and the moving dumbbell in Unity?

1. The Athletic Character & Rig

A stylized athletic lifter wearing a gym tank top and shorts. The arm is built with an anatomical hierarchy: `Shoulder → UpperArm → ElbowPivot → Forearm → Wrist → Hand`. Rotating the `ElbowPivot` automatically carries the forearm, hand, and dumbbell with it!

2. 3D Biceps Flexion Swell (Biofeedback)

When you curl in real life, your biceps bunches up! In Unity, a dynamic deformer script attaches to the biceps mesh. As your EMG activation increases, the 3D biceps mesh expands by **+45% in thickness** and **+50% in width**!

3. The Cast-Iron Hex Dumbbell

Modeled with a textured metallic knurled handle and two heavy hexagonal weights. It is parented directly to the hand bone so its position and rotation track the lifter's grip seamlessly.

4. Modern Gym & Studio Lighting

A sleek dark training studio featuring charcoal brick walls, dumbbell storage racks with rows of weights, a training bench with water bottle and towel, warm overhead key lights, and cool blue volumetric rim lights.

The Holographic "Ghost Target"

To guide the player, a 3D **Ghost Arm & Dumbbell** rendered in semi-transparent neon green shows the exact target pose:

- **Searching:** When outside the target angle, the ghost glows dim green (35% opacity).
- **In Zone:** The moment your arm enters within ±12° of the target, the ghost lights up into intense neon green with a pulsating **"IN ZONE"** badge floating above your hand!

Chapter 8: Designing the High-Tech Game UI & Experience

Inspired by modern fitness platforms (Apple Fitness+, Peloton), we designed a clean, futuristic HUD:

A **Top Navigation Bar with Glowing Chips:**
Features pill-shaped clickable chips: `HOLD / REPS` (challenge mode), `6 kg / NO LOAD` (weight setting), and `EMG | Filter IIR` (input source). It also shows the round status: `P1 • Round 2/3`.

B **Modern Floating Score Card:**
A dark glassmorphic card on the left displaying the current score (`2,840`), a golden multiplier pill (`x2` up to `x4`), the active streak count (`Streak: 6 in a row`), and the time spent on-target (`Time in zone: 74%`).

C **The Vertical Effort Thermometer:**
A neon-green bar on the right side of the screen displaying your live biceps effort from **0% to 100%**. A horizontal target line and highlighted bracket show you exactly where to hold your effort.

D **Circular Countdown Timer Ring:**
A radial fill ring with high-contrast digits showing the remaining seconds in the challenge (e.g., `23 s`) with smooth clockwise depletion.

E **Window Controls & Pause Menu (ESC):**
Includes modern desktop Minimize (`–`), Maximize (`□`), and Close (`×`) buttons. Pressing the **ESC** key pauses the game and opens a frosted menu with Resume, Fullscreen, Restart, and Quit options.

Chapter 9: The Code Machinery (Behind the Scenes)

Here is how the custom C# scripts work together inside Unity to power the entire experience:

Script	Role	Key Responsibilities
`UDPReceiver.cs`	**The Post Office**	Runs a background thread listening on UDP port 50557, parses incoming JSON packets, and safely transfers the activation value into the main game thread.
`ArmController.cs`	**The Puppeteer**	Interpolates between rest activation and required activation (0.60), rotates the elbow pivot between 0° and 140°, expands the 3D biceps mesh, and handles keyboard fallback (Up/Down arrow keys).
`GameManager.cs`	**The Game Director**	Controls the game loop for Hold Challenge (40°, 80°, 120°) and Reps Challenge, tracks scores, streaks, multipliers, updates UI meters, and handles hotkeys (`K` for input, `F` for filter, `M` for mode, `L` for load, `Space` to start).
`GameLogger.cs`	**The Scientist's Notebook**	Logs timestamp, elbow angle, target angle, in-zone status, and effort to CSV files on disk for research and course evaluations.
`WindowController.cs`	**The Window Manager**	Interacts with the Windows OS API to allow borderless dragging, window minimization, maximization toggle, and clean game shutdown.
`WeightliftingSceneBuilder.cs`	**The Master Architect**	An automated Unity Editor script that generates the entire 3D gym, lifter mesh, materials, lighting, cameras, canvases, and button wiring programmatically in seconds!

Chapter 10: The Grand Loop (Brain to Screen in 40 ms!)

When you put everything together, here is the complete journey of a single muscle curl:

1. **Your Brain:** You think: "*Lift the dumbbell to 80 degrees!*" Your brain fires nerve signals down your arm.
2. **The Muscle:** Biceps muscle cells depolarize, generating electrical sparks of ~500 µV.
3. **The Electrodes & BioRadio:** Sticky pads pick up the voltage; the BioRadio amplifies it and takes 1,000 samples per second.
4. **Bluetooth Transmission:** The samples fly through the air over Bluetooth to your computer.
5. **Python Signal Processing:** The IIR filter removes 50 Hz hum; rectification and RMS compute the clean linear envelope.
6. **Two-Point Scaling:** The calibration formula normalizes the envelope relative to your 6 kg rest and max levels, yielding an activation of 0.60.
7. **UDP Network Packet:** Python sends `{"activation": 0.60, ...}` to port 50557.
8. **Unity ArmController:** The elbow pivot rotates to 84°; the biceps mesh swells by +30%.
9. **Game Feedback:** The hand enters the holographic ghost target; **"IN ZONE"** glows neon green, the effort bar hits the target line, and your score starts climbing!

> **The Speed of Light:**
> How long did all 9 steps take? **Less than 40 milliseconds!** A human eye blink takes around 150 to 300 milliseconds. That means this entire loop from your brain to the 3D virtual world happens **five times faster than a single blink of an eye!**